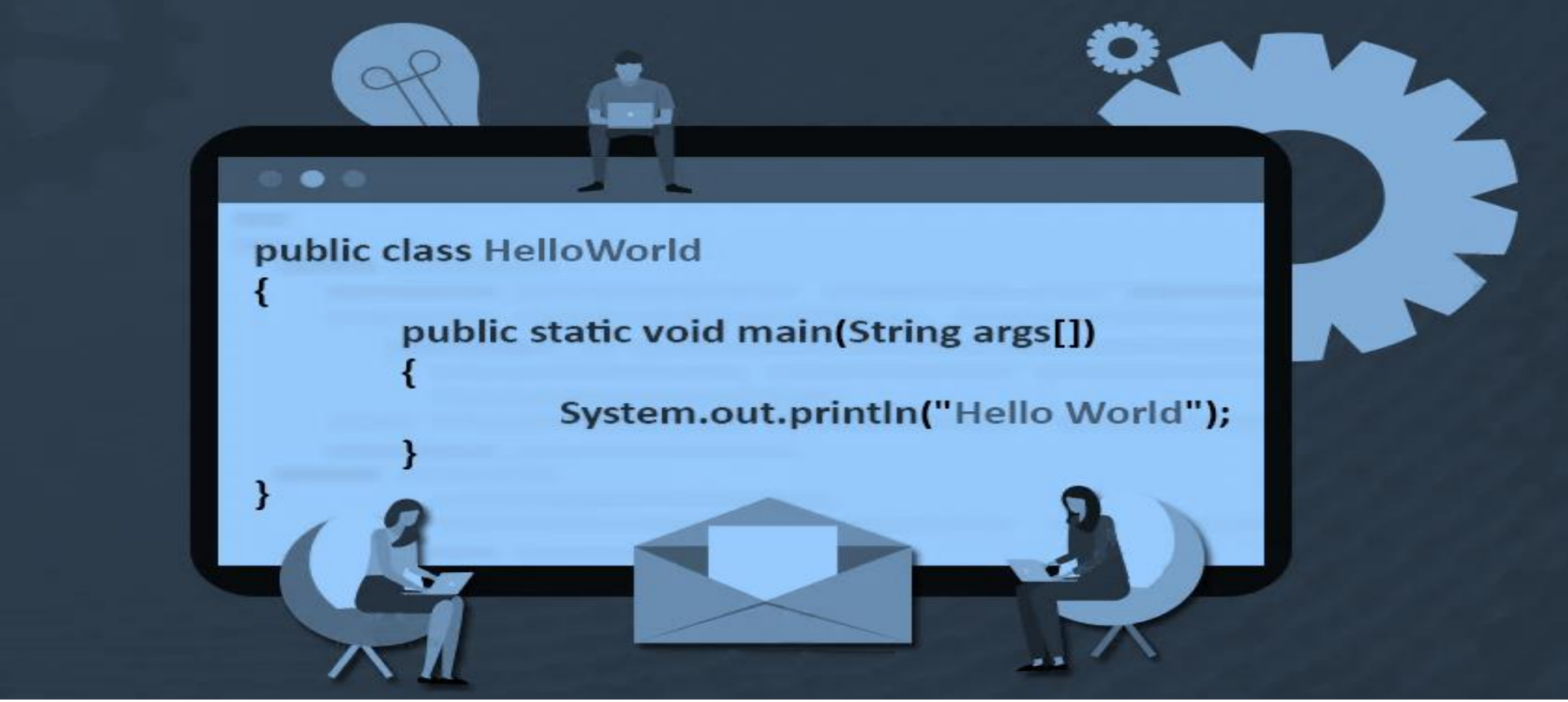


Java Programming

CPT111 – Lecture 1

Erick Purwanto

Xi'an Jiaotong-Liverpool University
西交利物浦大学

CPT111 Java Programming

Lecture 1

Simple Java Program, Data, Variables

Welcome!

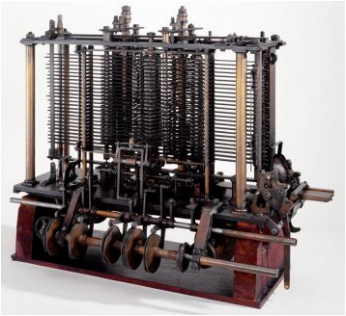
- Welcome to Lecture 1 !
  - it is recommended that you finish reading Lecture 0 first ...

- In this lecture we are going to learn to write and run a simple Java program
  - learn its elements
    - class
    - main method
    - statements
  - learn about variables
    - declaration
    - initialization
    - assignment

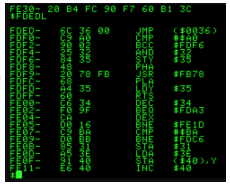
Learn to Program

- Apps are good when what they do is what you want

- Programming enables you to make a computer do anything you want
  - First programmer: Ada Lovelace
  - Published first published the first *algorithm* to the first general-purpose computer, the *Analytical Engine*



Programming Languange



- To tell computers what to do, need to learn a programming language
- Languages:
  - Machine language – easy for computers, hard for human
  - Natural language – easy for human, hard for computers
  - Programming language – in-between

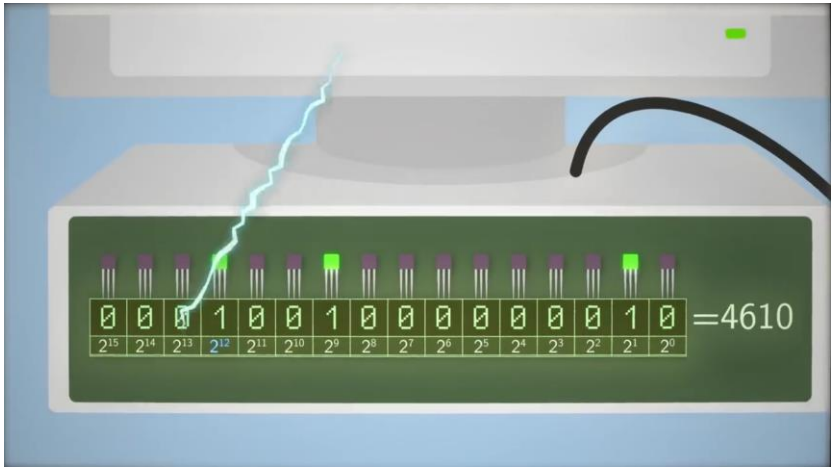
- Which programming language?
  - Let's first talk about how computer represent data and instructions ...

Binary, Bit and Bytes

- To a modern computer, everything is binary
  - 1 bit is 1 binary digit: 0 or 1 ; ON or OFF ; TRUE or FALSE
- Every bit we add doubles the total we can represent
  - 1 bit – 2
  - 2 bits – 4
  - 3 bits – 8
  - 4 bits – 16
  - ...
  - 8 bits – 256 – one Byte
- Mathematically: n bits yields 2^n patterns (2 to the n-th power)
  - https://www.wikihow.com/Convert-from-Decimal-to-Binary

The Universe is Hostile to Computers

- What happened if **one single bit** in a computer got flipped?
- Watch this video from Veritasium:
  - https://www.youtube.com/watch?v=AaZ_RSt0KP8



Bit, Bytes and Storage

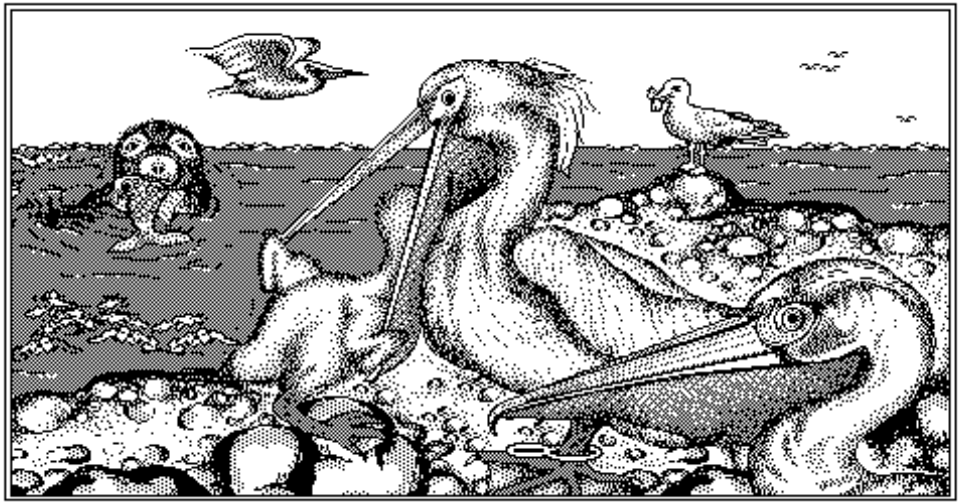
- More bits mean more potential storage
  - Longer instructions, more data
  - Better quality files

- 8 bits = 1 byte ≈ often 1 character, e.g. "a"
- 1024 bytes = 1 kilobyte ≈ several paragraphs
- 1024 kilobytes = 1 Megabyte ≈ about 200 pages of text
- 1024 Megabytes = 1 Gigabyte ≈ 256 MP3 files
- 1024 Gigabytes = 1 Terabyte ≈ 40 large movies

- Next, let's give some examples how things are stored/represented in binary!

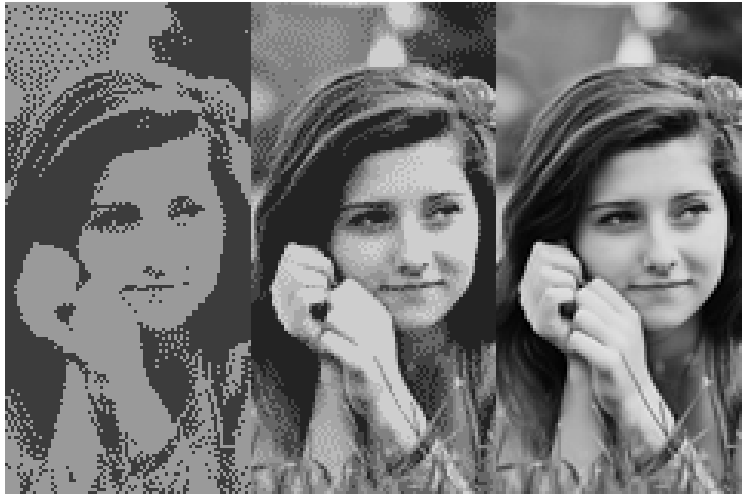
1-Bit Colour

- Black or white
  - with a high enough resolution some pretty good effects could be had by mixing black and white to get grey effects



8-Bit Colour – Grayscale

- 8-bit grayscale gave us decent photos – in "black and white" (which is not!)
  - 256 different colours possible



1 bit 2 bit 8 bit

8-Bit Colour

- Gives us 256 different colours
  - which 256 depends on the system

Computer Gaming Graphics (1)

- 1980s – Nintendo Entertainment System (NES)
  - 8 bit console
  - 54 colours

Computer Gaming Graphics (2)

- Early 1990s – Super Nintendo Entertainment System (SNES)
  - 16 bit console
  - 256 colours

Computer Gaming Graphics (3)

- Mid 1990s – Sony PlayStation
  - 32 bit console
  - 16,777,216 colours

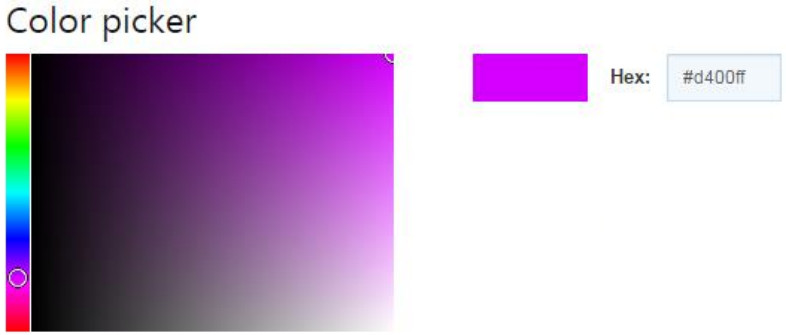
Hexadecimal

- Base 16 numbering system
  - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Represented with 0x__
  - i.e. 255 in decimal is 0xFF
- More readable than binary
  - decimal 121
  - binary 01111001
  - hex 0x79
- Stores more information than binary
  - hex and binary relate well
  - 1 hex digit can always be represented with 4 binary digits
  - no relationship like this between decimal and binary

decimal	hexadecimal	binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

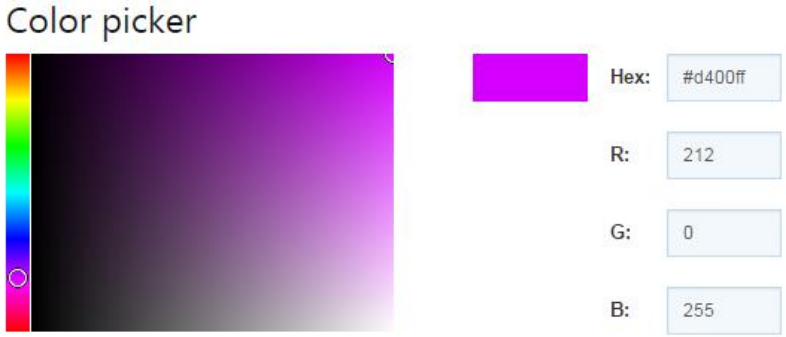
24-bit Colour (1)

- 24-bit colour means that there are 24 bits allocated for RGB channels
  - 8 bits for red, 8 bits for green, 8 bits for blue
- How to represent this colour purple?
  - rich purple? deep purple?
- Need a better representation
  - often use hexadecimal 0xd400ff
  - what does that mean?



24-bit Colour (2)

- Hex 0xd400ff
  - d4 – red bits
  - 00 – green bits
  - ff – blue bits



- Compact, as opposed to 24-bit value for this colour purple
  - 110101000000000011111111

ASCII – American Standard Code for Information Interchange

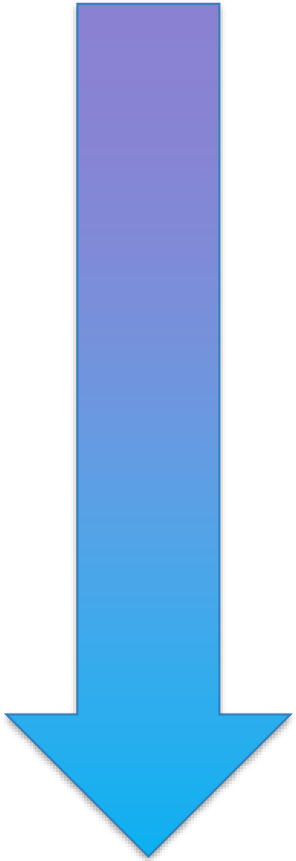
- Recall, computers can only store binary
  - includes characters
- ASCII is an encoding scheme that represents each character with a 7-bit value

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	|
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

Computer Instructions

- Computers also use binary for their instructions!
- However, we do not program in binary
  - this would be a nightmare to write and proofread
  - so many errors
- Programming languages are developed to help this
  - they can be low-level, or high-level

Low-Level to High-Level Language



- Binary: `10001101001110101001001001 11`

- Assembly:

```
TAD  I  Z  STPTR  / Get next character
           SNA    / Skip if non-zero (not end of string)
           HLT    / Else halt on zero (end of string)
           TLS    / Output the character in the AC to the teleprinter
```

- C, C++, C#, Python,
  and **Java**: `System.out.println("Hello World!");`

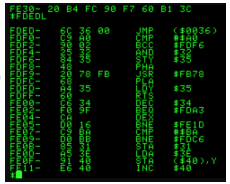
Assembly Language

- Low Level Languages
  - Very simple commands
  - Subtract, add, copy, delete
  - A command and then values
  - Interacts directly with memory and registers

- Assembly language
  - a low level language
  - easier to program in than binary
  - very simple commands,
    which can be easily translated to binary
  - however, still difficult to read and follow!
  - CPU instructions use this format

```
SUB     AX,AX
MOV     ES,AX
SUB     BH,BH
MOV     BL,INT_NUMBER
SHL     BX,1
```

High-Level / Programming Languange



- To tell computers what to do, need to learn a programming language
- Languages:
  - Machine language – easy for computers, hard for human
  - Natural language – easy for human, hard for computers
  - High-level / Programming language – in-between

- Which programming language?
  - Let's first talk about how computer represent data and instructions ...

In-Class Quiz

- What programming language(s) have you used previously?

  - ❑ Java
  - ❑ Python
  - ❑ C or C++
  - ❑ PHP or JS or similar

Java

- Created by James Gosling
- Modern object-oriented language
  - continuously improved since 1990s
- Widely used
  - mars rover, cell phones, web servers, ...

- Unfortunately, Java is verbose
  - using more words – compare with **Python**!

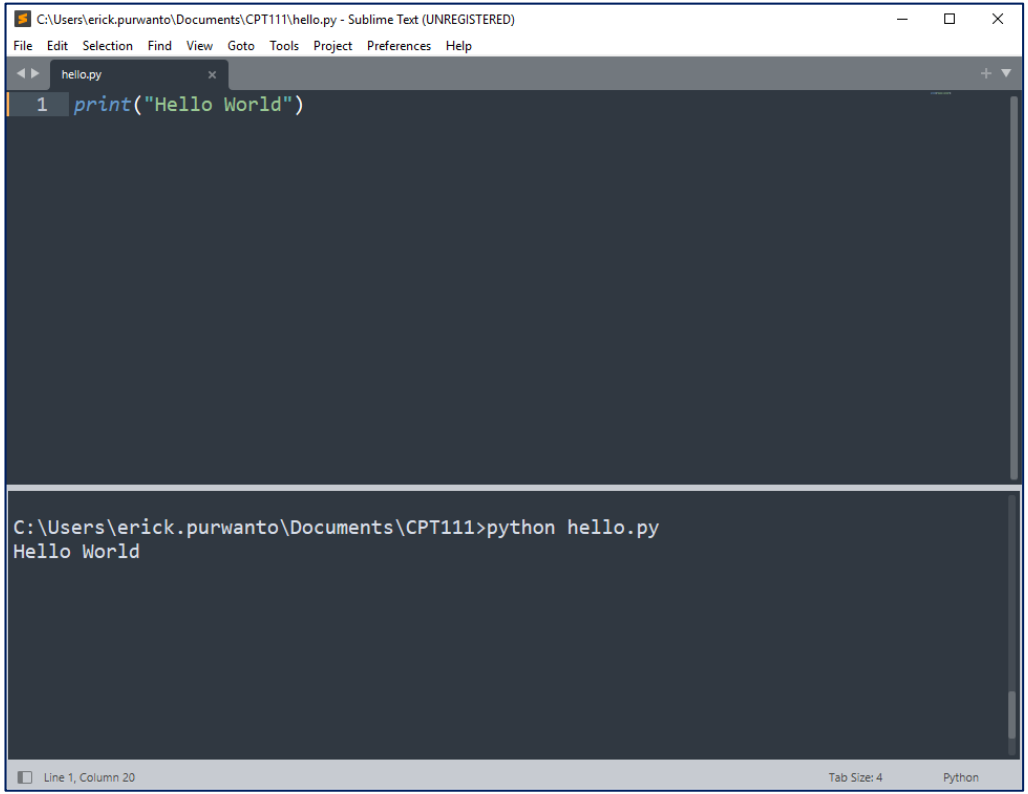
- More importantly, it is not about the language
  - the programming skills that you learn in this course will apply to many languages

First Program: Hello World

- Write a program that prints `"Hello World"` to the screen

First Program: Hello World

- Write a program that prints "Hello World" to the screen



First Program: Hello World

- Write a program that prints "Hello World" to the screen

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

Text Editor, Command Line Compilation and Execution

- We can use a text editor to write the source code
  - e.g. Notepad, Notepad++, Sublime Text

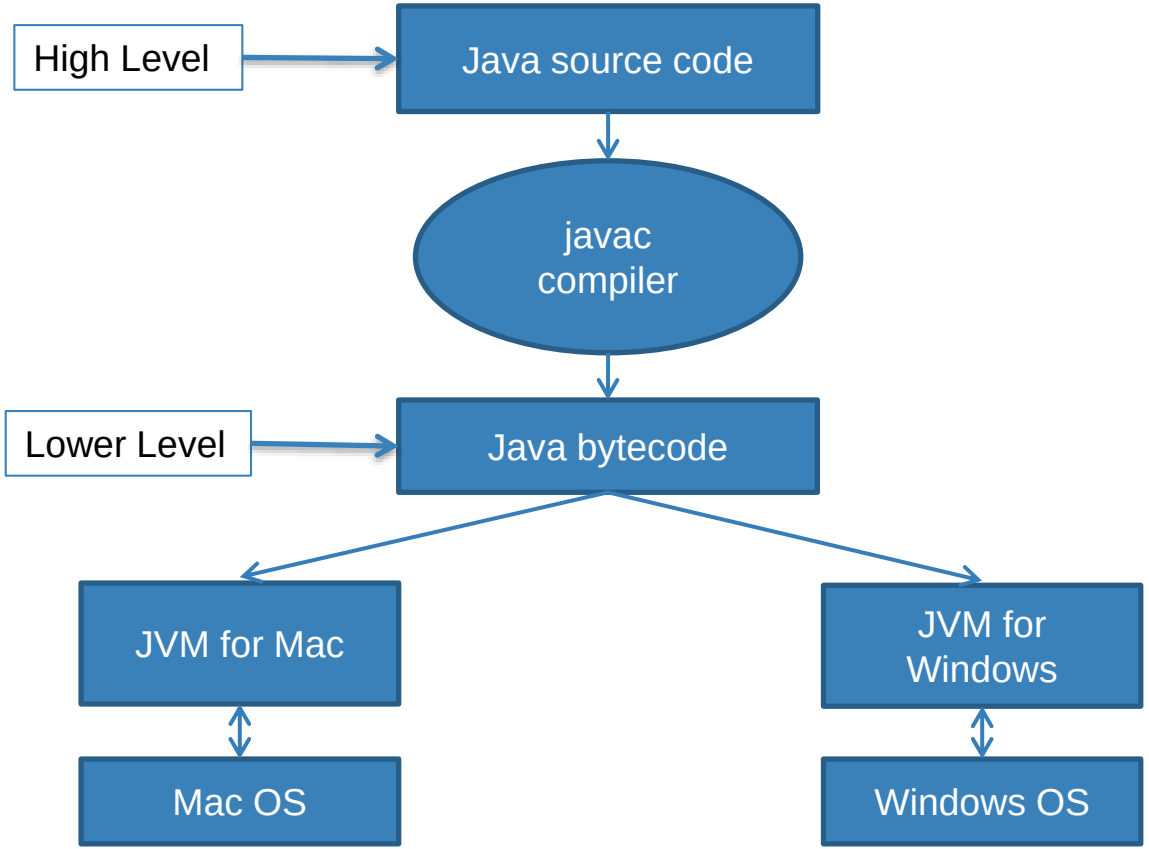
- We can compile and execute in command line
  - need to set the environment variables first
  - **`javac`** is used to compile programs
  - **`java`** is used to execute programs
  - we must always compile before execution

- Java source code is compiled into **Java bytecode**

Java Virtual Machine (JVM)

- JVM executes the Java bytecode
  - JVM is installed on various platform

- A Java class file is a file (with the **.class** filename extension) containing Java bytecode that can be executed on JVM



Elements of the Hello World Program

- What are the elements of the Hello World program?

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

class

main method

parameter

statement

braces

squared brackets

parentheses

semicolon

Object-Oriented Language

- Java is an Object-Oriented Language:
  - every Java file must contain a *class declaration*
  - all code is written inside a *class*
  - to run a Java program, need to define a *main method*

more OOP features in future lectures...

- Java's language features:
  - braces/curly brackets are used to denote the beginning and end of a block/a section of code
    - e.g. a class or method declaration
  - statements end with semicolons

- Violating one of the above will cause *syntax errors*

Variable

- Write a program that prints an integer variable num which stores an integer value 5

```
public class Hello {
    public static void main(String[] args) {
        int num = 5;
        System.out.println(num);
    }
}
```

variable

type

value

visualize the code execution in https://pythontutor.com/java.html

add some integer to the variable num before printing the result

expression

Static Typing

- Java is a statically typed language:
  - all variables, parameters, and methods must have a *declared type*
    - also called their *static type*
  - variables must be declared before use
  - that type can never change
  - expressions also have a type, e.g. `5 + 10` has type `int`
- Java compiler ensures type consistency
  - if types are inconsistent, the program will not compile
    - it is another example of *syntax error*
- Using an IDE, the compiler checks that all the types in your program are compatible *before* the program ever runs
  - a good thing, catch errors as early as possible

In-Class Quiz

- What is the output of the following program?

```
public class Hello {
    public static void main(String[] args) {
        int num = 5;
        num = num + num;
        System.out.println(num);
    }
}
```

- 0
- 5
- 10
- 55

what if you want to reuse the initial value of `num` in other computation?

Declaration, Initialization, Assignment

- Variables can be declared and initialized once, and assigned multiple times

```
public class Hello {
    public static void main(String[] args) {
        int num1 = 5;
        int num2;
        num2 = 10;
        int num = num1 + num2;
        System.out.println(num);
    }
}
```

declaration

initialization

assignment

Escape Characters

- Write a program that prints

  `"My favorite movie is "Lord of The Ring""` to the screen

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("My favorite movie
                            is \"The Lord of the Rings\"");
    }
}
```

- Special characters need to be escaped by adding the backslash \ symbol:
  - single quote \'
  - double quote \"
  - new line character \n
  - backslash \\

Thank you for your attention !

- In this lecture, you have learned:
  - how to compile and run a simple Java program
  - about the structure of a simple Java program
  - how to use variables to store data

- Please continue to go to Lab 1 this Thursday, and after that solve
  - CW1 #1.1, #1.2